

MIE312 Fluid Mechanics

Ch. 7 - Flow Past Immersed Body

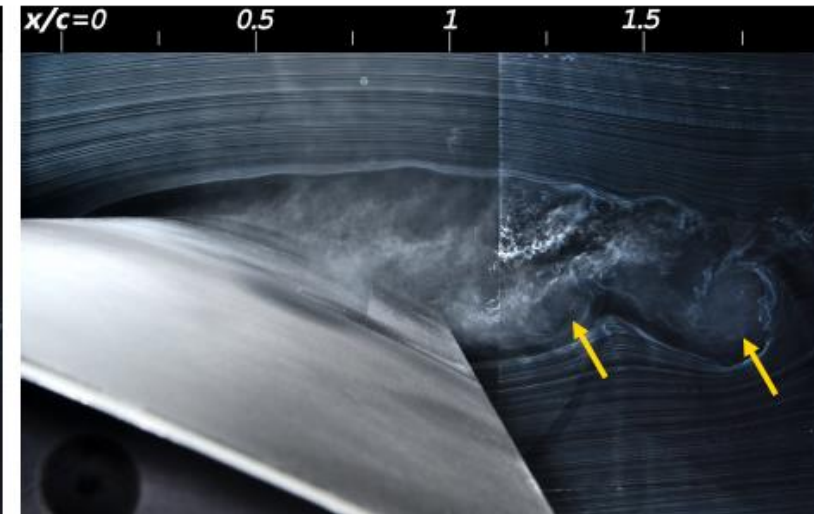
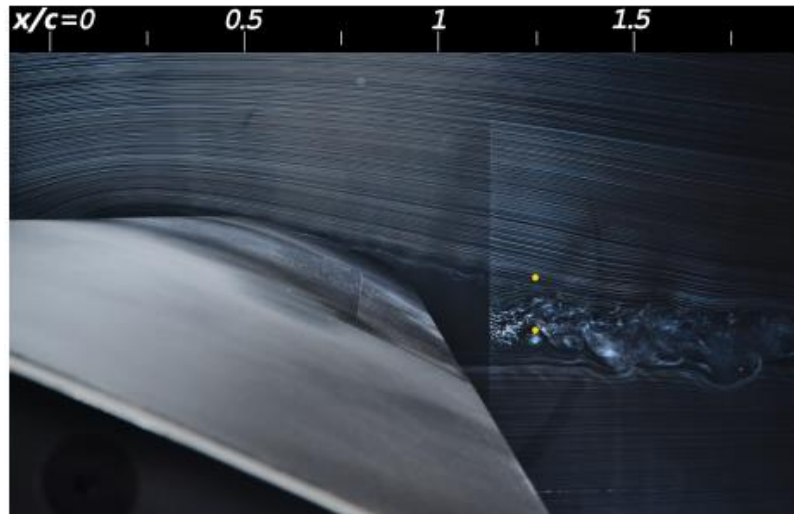
Howard Ho



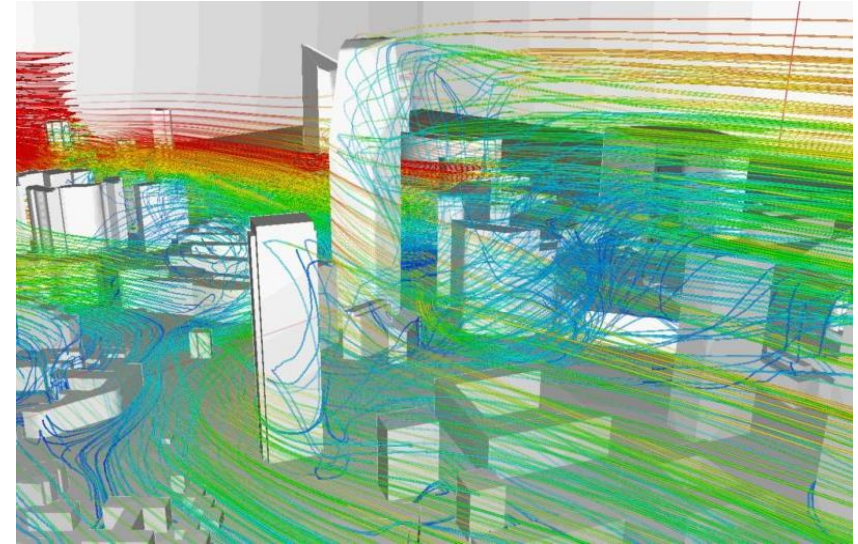
Learning Outcome

- Analysing problems in different types of external flow
- Understanding the concept of boundary layer
- Applying drag coefficient to find drag force on surfaces and bodies

Link to the summary video
<https://youtu.be/2YCtsl9X2UU>

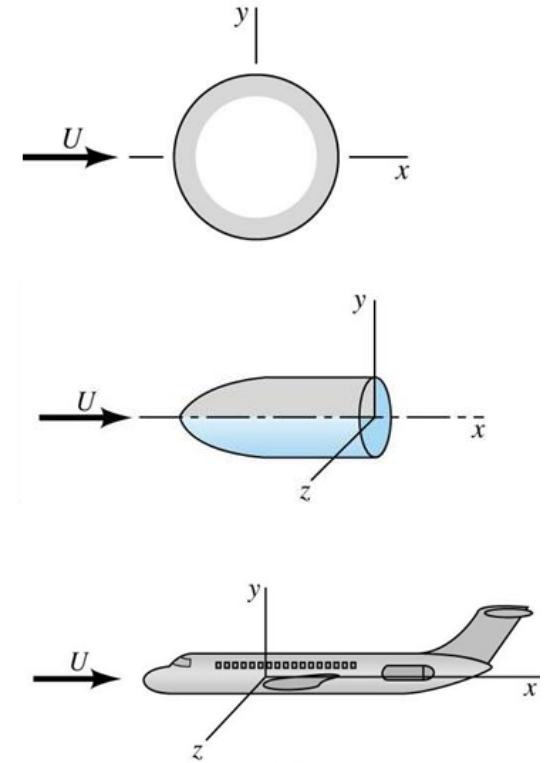


External Flow



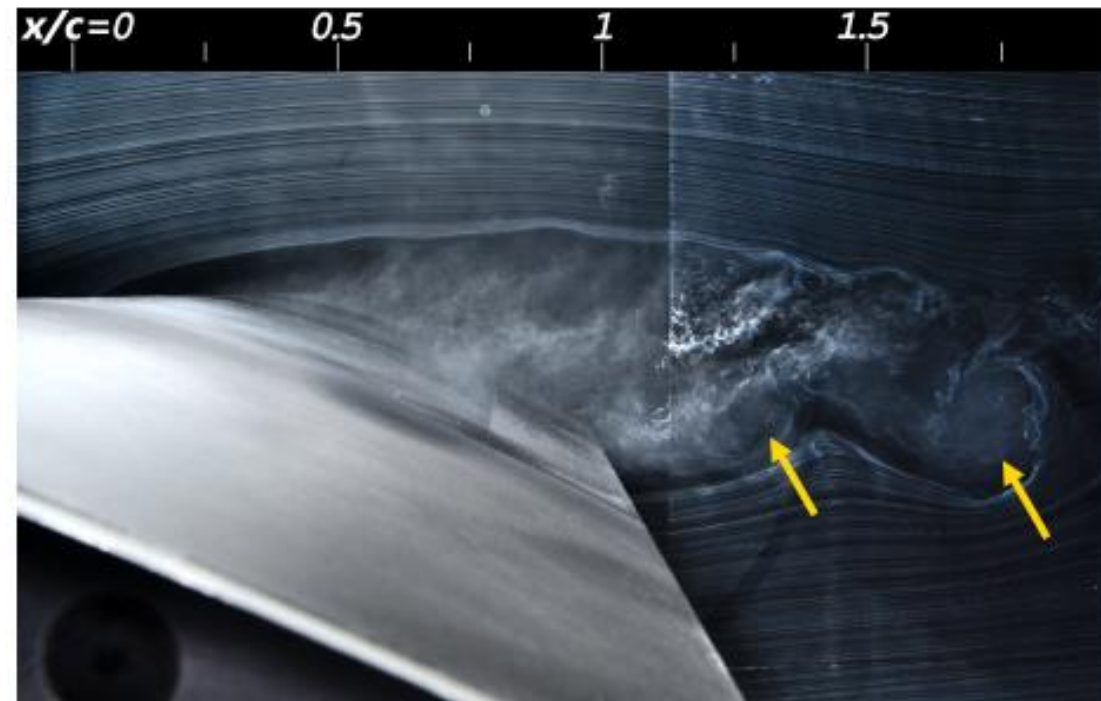
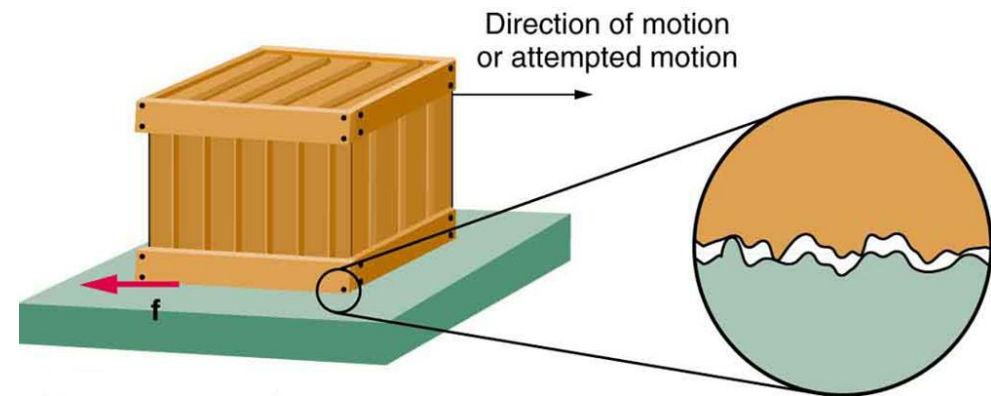
External Flow Type

- Two-Dimensional: infinitely long and fixed cross-section (into the page)
- Axisymmetric: shaped by axis of symmetry
- Three-Dimensional: can still be symmetric (to a plane rather than an axis)



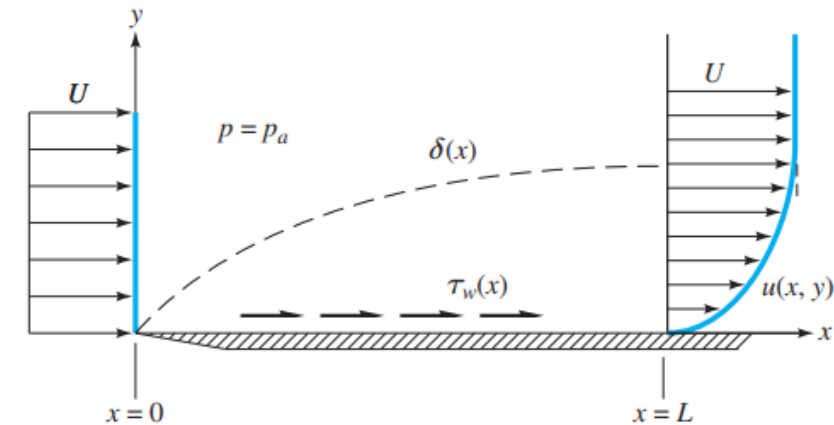
Viscous and Inviscid Flow

- When the flow is **inviscid**, the **shear (friction) is negligible**, this often occurs further away from walls
 - Standard Bernoulli Equation can be used (if all requirements are met)
- When we examine flow close to walls or in a wake, **significant shear occurs**, and thus the flow is **viscous**
 - Such as pipe flow from Ch.6, Energy Transport Equation (RTT) is needed



Boundary Layer

- **Boundary layer (BL)** is a thin region of fluid near a solid surface where the effects of viscosity (fluid friction) are significant. The friction causes flow near the surface to slow down (to basically stationary if you look very close to the surface)

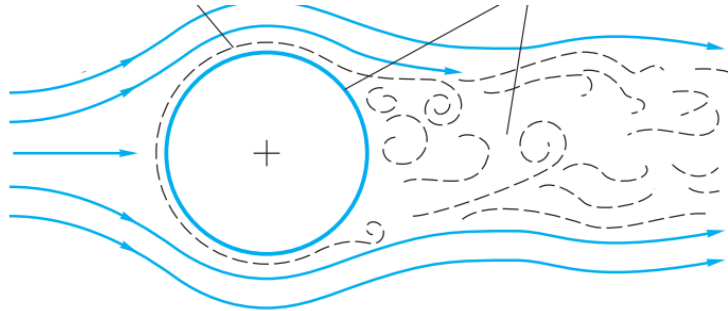
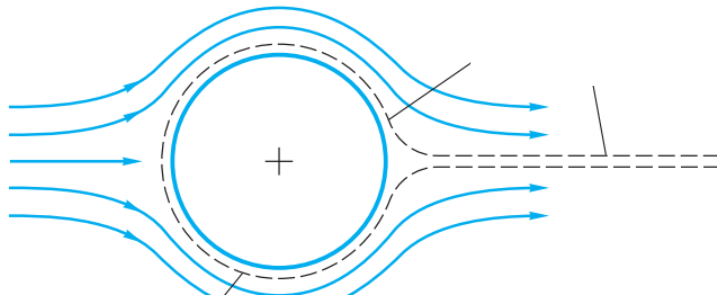
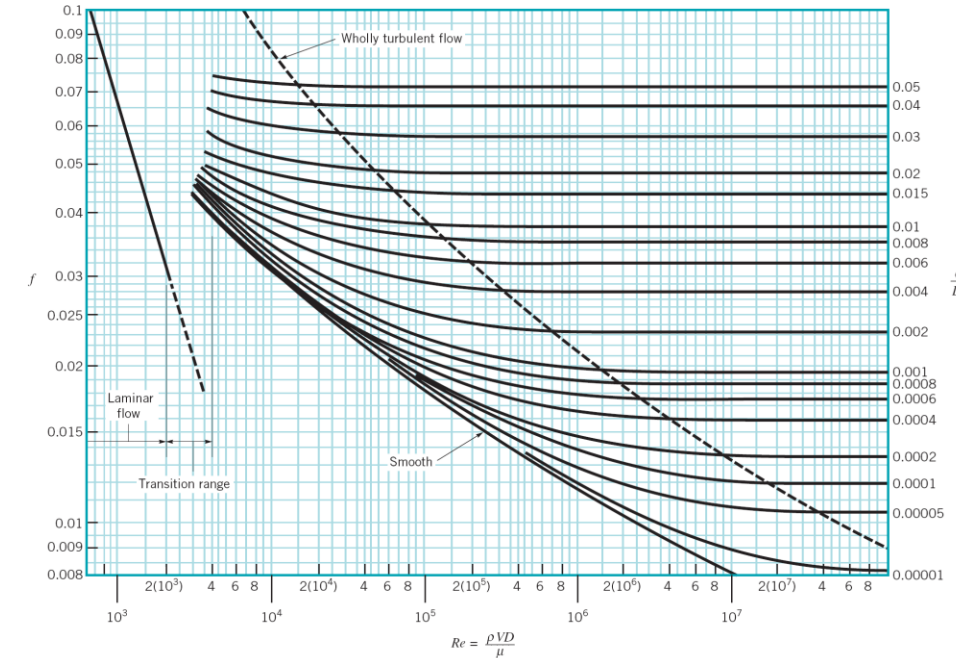


Reynolds Number (Flat Plate)

- Reynolds number for flow over flat plate can be calculated with

$$Re_L = \frac{\rho UL}{\mu}$$

- Laminar $Re < 10^6$
 - Orderly flow (clean, neat and reversible)
- Turbulence $Re > 10^6$
 - Chaotic (imagine chaos theory etc.)
- Technically, the transition is not a “light switch”
 - Transitional Region (treat that as a third flow regime)
 - Critical transition (assume it goes directly from laminar to turbulent at Re_{crit})



Transitional flow is a beast in real life, we tend to avoid or minimize it for most applications



Drag Force

- Drag Force F_D can be calculated as:

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

where,

C_D = Drag coefficient (calculate or given in table/chart) []

ρ = Density of fluid [kg/m^3]

V = Free-stream (in-coming) velocity [m/s]

A = Area [m^2]



Fun fact, the C_D equation is constructed based on Buckingham Pi Theorem (Ch.5):

$$\Pi_1 = f(\Pi_2)$$

$$C_D = f(Re)$$

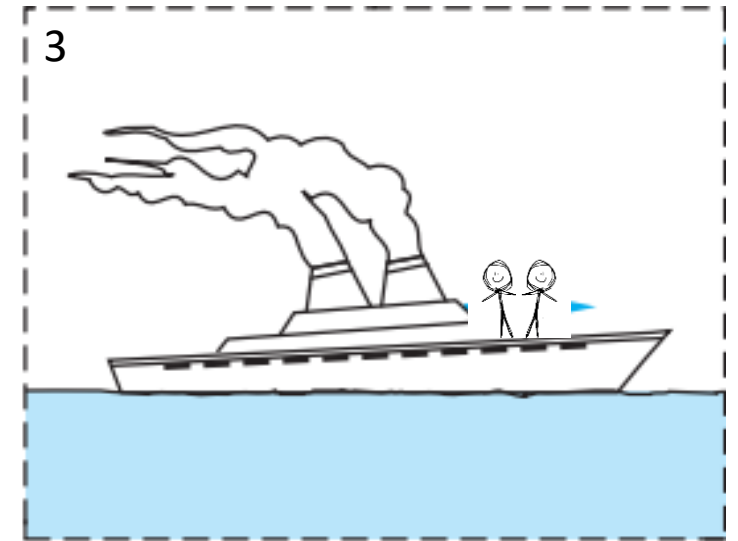
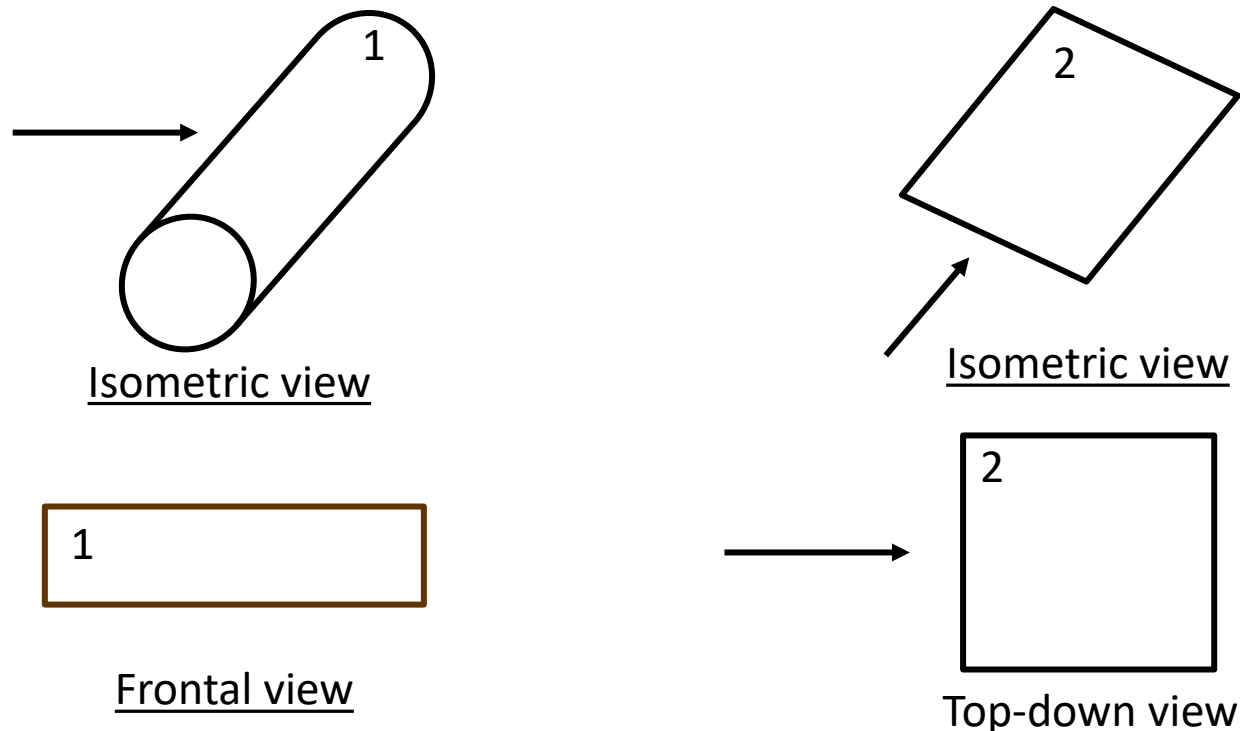
$$\frac{2F_D}{\rho V^2 A} = f\left(\frac{\rho UL}{\mu}\right)$$



Which Area to Pick?

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

- The choice of A (Area) can be tricky, there are 3 main types:
 - 1. Frontal area: For thick, bluff bodies, such as spheres, cylinders.
 - 2. Planform area (Top-down view): For wide, flat bodies such as flat plates and airfoils.
 - 3. Wetted area: For surface ships and barges.



Steps To Solve Drag Problems

1. Calculate Re , so we know if flow is laminar or turbulent ($Re = \frac{\rho UL}{\mu} = \frac{UL}{\nu}$)

2. Calculate or find C_D if not given (can also find BL height δ at this step)

- Laminar Plate ($Re < 10^6$): $C_D = \frac{1.328}{Re^{1/2}} ; \frac{\delta}{x} = \frac{5}{Re^{1/2}}$
- Turbulent Plate ($Re > 10^6$): $C_D = \frac{0.031}{Re^{1/7}} ; \frac{\delta}{x} = \frac{0.16}{Re^{1/7}}$

• Transitional:



$$C_D \approx \begin{cases} \frac{0.031}{Re_L^{1/7}} - \frac{1440}{Re_L} & Re_{trans} = 5 \times 10^5 \\ \frac{0.031}{Re_L^{1/7}} - \frac{8700}{Re_L} & Re_{trans} = 3 \times 10^6 \end{cases}$$

3. Find the area type (frontal or top-down)

4. Calculate F_D with the drag force equation ($F_D = \frac{1}{2} C_D \rho V^2 A$) **X2 afterwards if needed**

Question 1

7.26 Consider laminar flow past the square-plate arrangements in the figure below. Compared to the drag of a single plate (1), how much larger is the drag of four plates together as in configurations (a) and (b)? Explain your results.

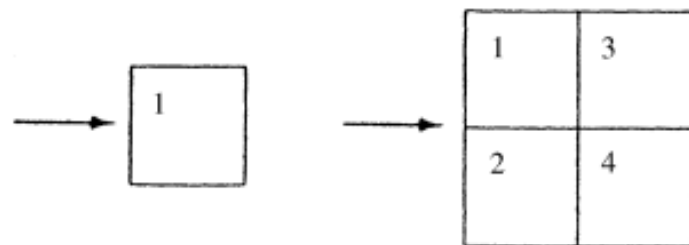


Fig. P7.26 (a)

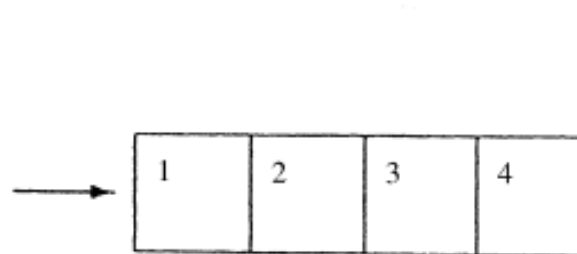
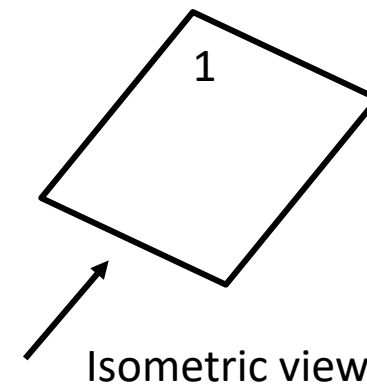
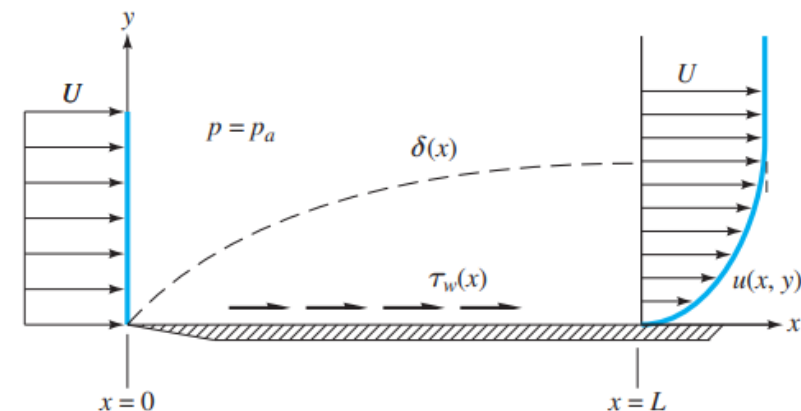


Fig. P7.26 (b)

Link to the example problems solving video:
<https://youtu.be/LA-n139k3SY>



4 Steps Guide

- 1) Find Re
- 2) Find C_D
- 3) Find A
- 4) Find F_D (x2 maybe?)

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

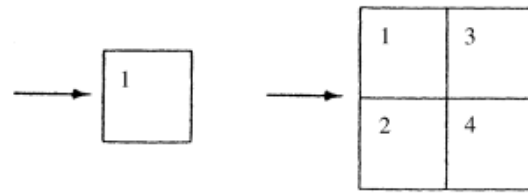


Fig. P7.26 (a)

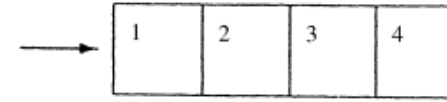


Fig. P7.26 (b)

4 Steps Guide

- 1) Find Re
- 2) Find C_D
- 3) Find A
- 4) Find F_D (x2 maybe?)

$$F_D = \frac{1}{2} C_D \rho V^2 A$$

Question 2

A hydrofoil 1.2 ft long and 6 ft wide is placed in a seawater flow of 40 ft/s, with $\rho = 1.99$ slugs/ft³ and $\nu = 0.000011$ ft²/s. (a) Estimate the boundary layer thickness at the end of the plate. Estimate the friction drag for (b) turbulent smooth-wall flow from the leading edge, (c) laminar turbulent flow with $Re_{\text{trans}} = 5 \times 10^5$

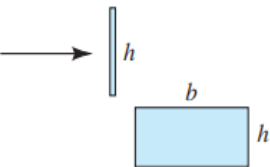


4 Steps Guide

- 1) Find Re
- 2) Find C_D
- 3) Find A
- 4) Find F_D (x2 maybe?)

Question 3

Table 7.3 Drag of Three-Dimensional Bodies at $Re \geq 10^4$

Body	Ratio	C_D based on frontal area
Rectangular plate: 	b/h 1	1.18
	5	1.2
	10	1.3
	20	1.5
	∞	2.0

4 Steps Guide

- 1) Find Re
- 2) Find C_D
- 3) Find A
- 4) Find F_D (x2 maybe?)

7.88 A pickup truck has a clean drag-area $C_D A$ of 35 ft^2 . Estimate the horsepower required to drive the truck at 55 mi/h (a) clean and (b) with the 3- by 6-ft sign in Fig. P7.88 installed if the rolling resistance is 150 lbf at sea level.

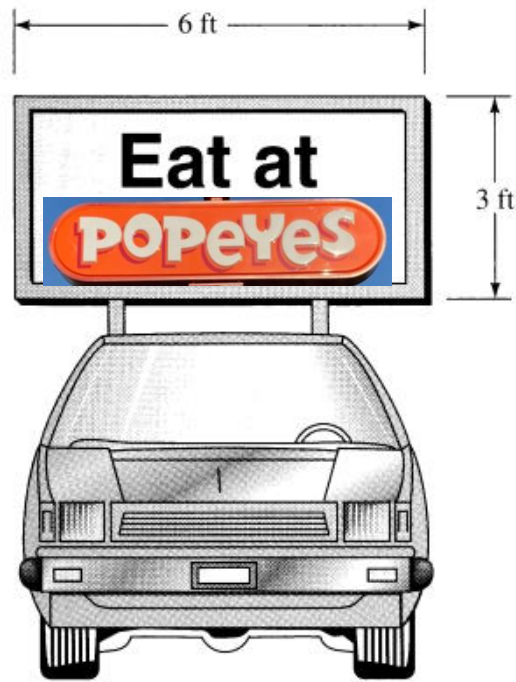


Fig. P7.88

Question 1

7.26 Consider laminar flow past the square-plate arrangements in the figure below. Compared to the drag of a single plate (1), how much larger is the drag of four plates together as in configurations (a) and (b)? Explain your results.

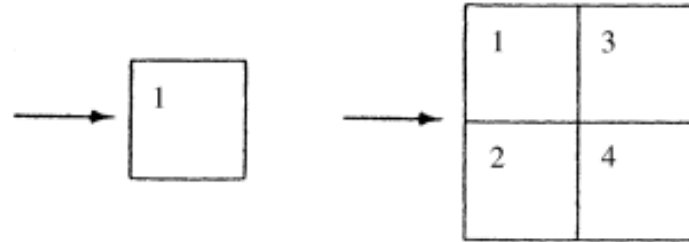


Fig. P7.26 (a)

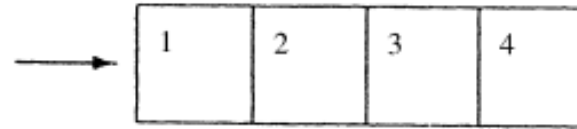


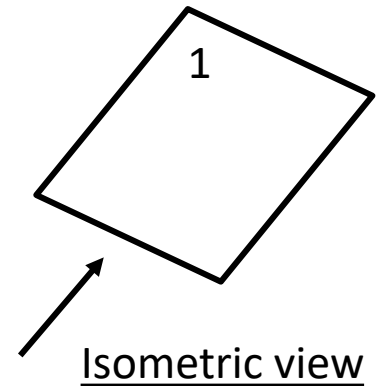
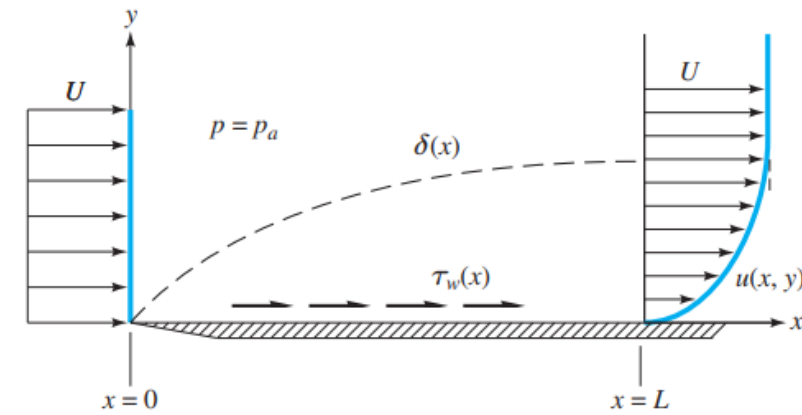
Fig. P7.26 (b)

Solution: The laminar formula $C_D = 1.328/\text{Re}_L^{1/2}$ means that $C_D \propto L^{-1/2}$. Thus:

$$(a) F_a = \frac{\text{const}}{\sqrt{2L_1}}(4A_1) = \sqrt{8}F_1 = \mathbf{2.83F_1} \quad \text{Ans. (a)}$$

$$(b) F_b = \frac{\text{const}}{\sqrt{4L_1}}(4A_1) = \mathbf{2.0F_1} \quad \text{Ans. (b)}$$

The plates near the trailing edge have less drag because their boundary layers are thicker and their wall shear stresses are less. These configurations do **not** quadruple the drag.



Question 2

EXAMPLE 7.4

A hydrofoil 1.2 ft long and 6 ft wide is placed in a seawater flow of 40 ft/s, with $\rho = 1.99$ slugs/ft³ and $\nu = 0.000011$ ft²/s. (a) Estimate the boundary layer thickness at the end of the plate. Estimate the friction drag for (b) turbulent smooth-wall flow from the leading edge, (c) laminar turbulent flow with $Re_{trans} = 5 \times 10^5$, and (d) turbulent rough-wall flow with $\varepsilon = 0.0004$ ft.

Solution

The Reynolds number is

$$Re_L = \frac{UL}{\nu} = \frac{(40 \text{ ft/s})(1.2 \text{ ft})}{0.000011 \text{ ft}^2/\text{s}} = 4.36 \times 10^6$$

Thus the trailing-edge flow is certainly turbulent. The maximum boundary layer thickness would occur for turbulent flow starting at the leading edge. From Eq. (7.42),

$$\frac{\delta(L)}{L} = \frac{0.16}{(4.36 \times 10^6)^{1/7}} = 0.018$$

or $\delta = 0.018(1.2 \text{ ft}) = 0.0216 \text{ ft}$ Ans. (a)

This is 7.5 times thicker than a fully laminar boundary layer at the same Reynolds number.

For fully turbulent smooth-wall flow, the drag coefficient on one side of the plate is, from Eq. (7.45),

$$C_D = \frac{0.031}{(4.36 \times 10^6)^{1/7}} = 0.00349$$

Then the drag on both sides of the foil is approximately

$$D = 2C_D(\frac{1}{2}\rho U^2)bL = 2(0.00349)(\frac{1}{2})(1.99)(40)^2(6.0)(1.2) = 80 \text{ lbf} \quad \text{Ans. (b)}$$

Part d uses a special chart to find C_D , similar to the moody chart but for plates (as if one moody chart isn't enough)



With a laminar leading edge and $Re_{trans} = 5 \times 10^5$, Eq. (7.49a) applies:

$$C_D = 0.00349 - \frac{1440}{4.36 \times 10^6} = 0.00316$$

The drag can be recomputed for this lower drag coefficient:

$$D = 2C_D(\frac{1}{2}\rho U^2)bL = 72 \text{ lbf} \quad \text{Ans. (c)}$$

Finally, for the rough wall, we calculate

$$\frac{L}{\varepsilon} = \frac{1.2 \text{ ft}}{0.0004 \text{ ft}} = 3000$$

From Fig. 7.6 at $Re_L = 4.36 \times 10^6$, this condition is just inside the fully rough regime. Equation (7.48b) applies:

$$C_D = (1.89 + 1.62 \log 3000)^{-2.5} = 0.00644$$

and the drag estimate is

$$D = 2C_D(\frac{1}{2}\rho U^2)bL = 148 \text{ lbf} \quad \text{Ans. (d)}$$

This small roughness nearly doubles the drag. It is probable that the total hydrofoil drag is still another factor of 2 larger because of trailing-edge flow separation effects.

Question 3

7.88 A pickup truck has a clean drag-area $C_D A$ of 35 ft^2 . Estimate the horsepower required to drive the truck at 55 mi/h (a) clean and (b) with the 3- by 6-ft sign in Fig. P7.88 installed if the rolling resistance is 150 lbf at sea level.

Solution: For sea-level air, take $\rho = 0.00238 \text{ slug/ft}^3$ and $\mu = 3.72\text{E-}7 \text{ slug/ft}\cdot\text{s}$. Convert $V = 55 \text{ mi/h} = 80.7 \text{ ft/s}$. Calculate the drag without the sign:

$$F = F_{\text{rolling}} + C_D A \frac{\rho}{2} V^2$$

$$= 150 + 35(0.00238/2)(80.7)^2 \approx 421 \text{ lbf}$$

$$\text{Horsepower} = (421)(80.7) \div 550 \approx \mathbf{62 \text{ hp}} \text{ (clean) } \textit{Ans. (a)}$$

With a sign added, $b/h = 2.0$, read $C_D \approx \mathbf{1.19}$ from Table 7.3. Then

$$F = 421_{\text{clean}} + 1.19 \left(\frac{0.00238}{2} \right) (80.7)^2 (6)(3) \approx 587 \text{ lbf},$$

$$\text{Power} = FV \approx \mathbf{86 \text{ hp}} \textit{ Ans. (b)}$$

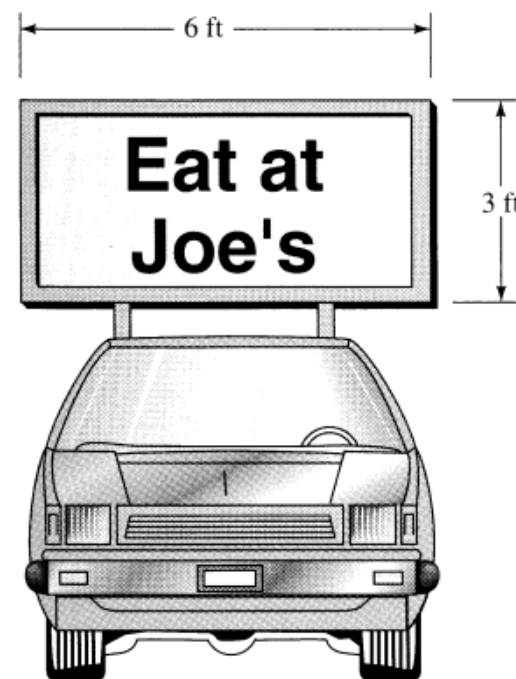


Fig. P7.88

Question 4*



7.36 A ship is 125 m long and has a wetted area of 3500 m². Its propellers can deliver a maximum power of 1.1 MW to seawater at 20°C. If all drag is due to friction, estimate the maximum ship speed, in kn.

Solution: For seawater at 20°C, take $\rho = 1025 \text{ kg/m}^3$ and $\mu = 0.00107 \text{ kg/m}\cdot\text{s}$. Evaluate

$$\text{Re}_L = \frac{\rho UL}{\mu} = \frac{1025V(125)}{0.00107} \text{ (surely turbulent), } C_D = \frac{0.031}{\text{Re}_L^{1/7}} = \frac{0.00217}{V^{1/7}}$$

$$\text{Power} = FV = \left[\frac{0.0217}{V^{1/7}} \left(\frac{1025}{2} \right) V^2 (3500) \right] V = 1.1\text{E}6 \text{ watts, or } V^{20/7} \approx 282.0$$

Solve for $V = 7.2 \text{ m/s} \approx \mathbf{14 \text{ knots.}}$ *Ans.*

Note: This is a supplementary example on how to deal with wetted area problems

Contact



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Linked-in: <https://www.linkedin.com/in/howard-haonan-ho/>

Email/ MS Teams: howard.ho@mail.utoronto.ca

Office: MC 66
